

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2461223.885 +/- 0.0024 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.083 +/- 0.041
Transit depth (Rp/Rs)^2	0.69 +/- 0.68 [%]
Orbital Inclination (inc)	82.26 +/- 2.51
Airmass coefficient 1 (a1)	0.117081 +/- 1.2e-05
Airmass coefficient 2 (a2)	1.73 +/- 0.98
Scatter in the residuals of the lightcurve fit is	0.014 %
Best Comparison Star	#3 - [430, 23]
Optimal Aperture	0.75
Optimal Annulus	4.75
Transit Duration (day)	0.061 +/- 0.057

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.083 $\pm$ 0.041 vs 0.1144 (deviation 0.77 σ)
Duration fit vs published	0.061 d vs 0.1471 d (deviation 1.51 σ)
Mid-time O-C	4.6 min
Depth SNR	2.0
Residual scatter	0.014 %

Light curve

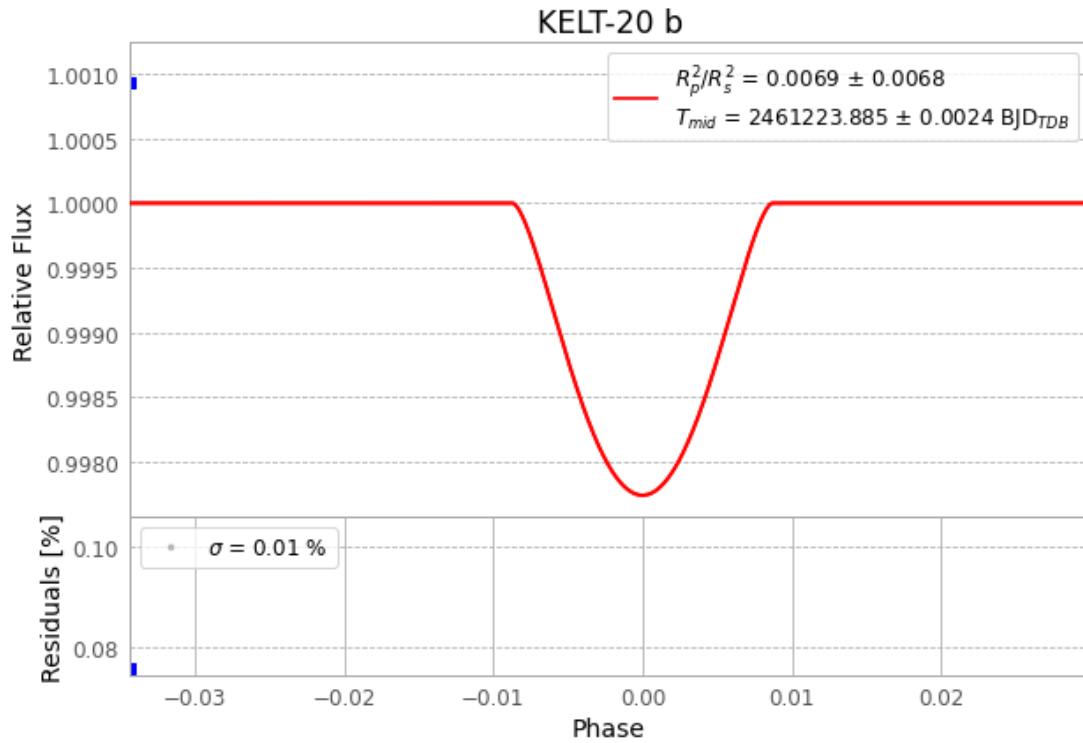


Figure 1 — FinalLightCurve_KELT-20 b_2026-07-02.png (SHA-256 5dcecc015ec0eb57...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [487, 396], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 c7ebcaa9ac1f0777...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13

Data provenance

111 FITS frames (73 MB), KELT-20260702061628.FITS → KELT-20260702122345.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: exotic-kelt20.log (aed0f17e8c12...), FinalLightCurve_KELT-20 b_2026-07-02.png (5dcecc015ec0...), FinalLightCurve_KELT-20 b_2026-07-02.csv (aa5b61c46041...)